# PHARMACY INVENTORY SYSTEM

GROUP 10

Before solutions can be implemented, the problem must be fully understood. This initial phase involves comprehensive assessment of existing inventory workflows, data collection on stockouts and expired medications, and stakeholder engagement, establishing the blueprint for meaningful improvement.

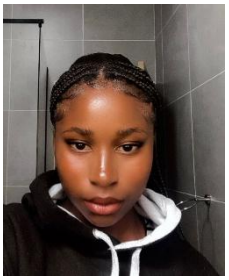
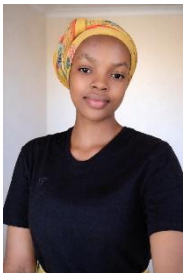
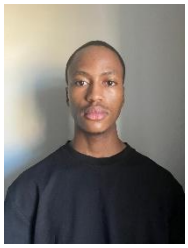
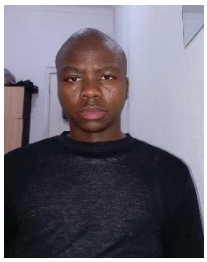
# Table of Contents


| | |
|---|---|
| Group Information | 2 |
| Photos of the Group Members | 3 |
| Business Rules | 4 |
| Entities and their primary keys, foreign keys and attributes | 5-6 |
| ER Diagram | 7 |
| Relationships | 8-11 |
| Diagram Normalization | 11 |
| Features | 12 |
| Logical Data Model | 13-15 |

# Group Information

| Name and Surname | Student Number | Contact Information | Member Involvement |
|---|---|---|---|
| Mthimunye Sakhile (Group Leader) | 45224706 | 45224706@mynwu.ac.za 0768622463 | Participated |
| Mashao Keabetswe Beyonce | 45413657 | 45413657@mynwu.ac.za 0729704991 | Participated |
| Sedze Tanaka Bruce | 52827097 | 52827097@mynwu.ac.za 0616577525 | Participated |
| Mokgohloa Mpho | 50591509 | 50591509@mynwu.ac.za 0623206954 | Participated |
| Shibambo Karabo | 46221042 | 46221042@mynwu.ac.za 0729482657 | Participated |
| Mbele Paul Mpho | 53593634 | 53593634@mynwu.az.za 0608627044 | Participated |
| Konyana Lethabo | 46967494 | 46967434@mynwu.ac.za 0662420677 | Participated |
| Lamola Charmaine | 40624102 | 40624102@mynwu.ac.za 0763034700 | Participated |

# Photos of group members:

| | | | |
|---|---|---|---|
| <br>Mthimunye Sakhile<br>(Group Leader) | <br>Sedze Tanaka Bruce |  <br>Mashao Keabetswe Beyonce |  <br>Shibambo Karabo |
| <br>Mbele Paul Mpho |  <br>Konyana Lethabo | <br>Lamola Charmaine |  <br>Mokgohloa Mpho |

# BUSINESS RULES

1. A user account must be created before assigning any role, it can have one role at the time but cannot be both a pharmacist and patient simultaneously in the system.
2. Each user must have a unique email address and password for system authentication
3. A supplier can supply one or many products.
4. A product is supplied by exactly one supplier.
5. A product can have one or many stock batches.
6. A stock batch belongs to exactly one product.
7. A patient can have one or many prescriptions.
8. A prescription belongs to exactly one patient.
9. A pharmacist can dispense one or many prescriptions.
10. A prescription is dispensed by exactly one pharmacist.
11. A prescription can contain one or many products.
12. A product can appear on one or many prescriptions.
13. A stock controller can receive one or many stock batches into the system.
14. A stock batch is received by exactly one stock controller.
15. A sales transaction can include one or many products.
16. A product can be sold in one or many sales transactions.
17. A controlled substance log entry must include product, pharmacist, patient, and batch.
18. A purchase order is placed with exactly one supplier.
19. A supplier can have one or many purchase orders.
20. Each batch must have expiry data, and the system must monitor it.
21. Whenever stock comes in or goes out, the quantity must be updated.
22. A sales transaction can contain one or many products via sales lines and a sales line belongs to exactly one purchase order and exactly one product.
23. A purchase order can request one or many products via purchase sales lines.
24. A purchase order line belongs to exactly one order and exactly one product

# ENTITIES AND THEIR PRIMARY KEYS, FOREIGN KEYS AND ATTRIBUTES

**USER**

- USER_ID(Primary key), first_name, last_name, date_of_birth, role, email, phone_number, address, password, gender, country, reg date.

**SUPPLIER**

- SUPPLIER_ID(Primary Key), supplier_name, email, address, phone_number, status, payment_term, license number.

**PRODUCT**

- PRODUCT_ID(Primary Key), product_name, description, dosage, category, price, SUPPLIER_ID(Foreign Key), expiry date, reorder quantity, storage req.

**STOCK BATCH**

- BATCH_ID(Primary Key), PRODUCT_ID(Foreign Key), manuf_ate, storage_conditions, quantity, batch_number, expiry_date, unit_cost, CONTROLLER_ID, received_date.

**PURCHASE ORDER**

- ORDER_ID(Primary Key), order_date, SUPPLIER_ID(Foreigen Key), delivery_date, status, quantity_ordered, unit_cost.

**PURCHASE ORDER LINE**

- Composite Primary Key, Foreign Key (ORDER_ID+ PRODUCT_ID).

**PATIENT**

- PATIENT_ID(Primary Key), USER_ID(fk), passport/id_number.

**PRESCRIPTION**

- PRESCRIPTION_ID(Primary Key), date_issued, PATIENT_ID(fk), PHARMAACIST_ID(Foreign Key), doctor_name, clinic/hospital_name.

**PRESCRIPTION LINE**

- Composite Primary Key, Foreign Key (PRODUCT_ID + PRESCRIPTION_ID), dosage_description, quantity.

**PHARMACIST**

- PHARMACIST_ID(Primary Key), USER_ID(Foreign Key), license_number, status.

**STOCK CONTROLLER**

- CONTROLLER_ID(Primary Key), USER_ID(Foreign Key)

**TRANSACTION**

- TRANSACTION_ID(Primary Key), transaction_date, type, PATIENT_ID(Foreign Key), PHARMACIST_ID(Foreign Key)

**CONTROLLED LOG**

- LOG_ID(Primary Key), log_date, PRODUCT_ID(Foreign Key), PHARMACIST_ID (Foreign Key), PATIENT_ID(Foreign Key), status, action_type, quantity, BATCH_ID (Foreign Key).

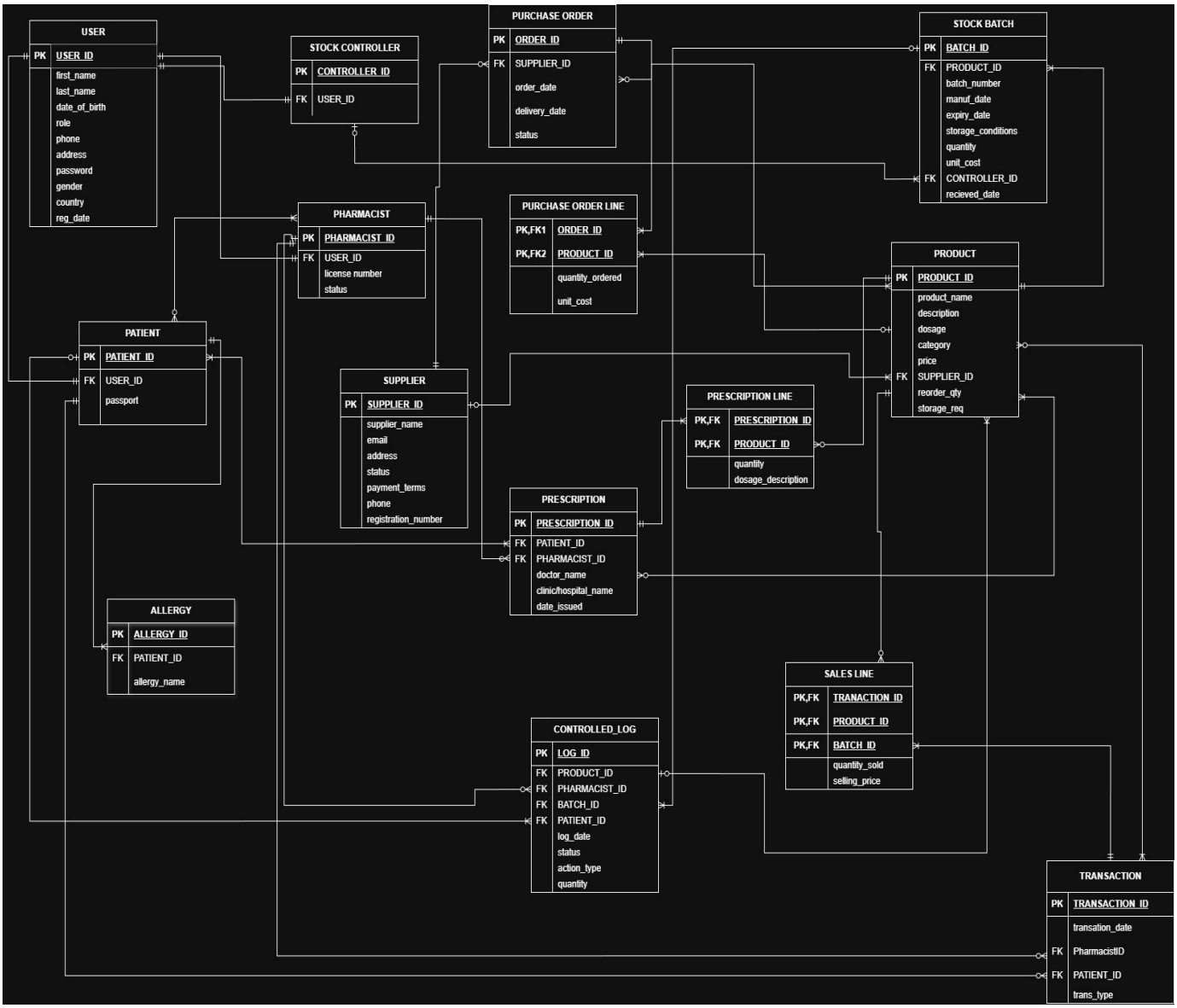
**SALES LINE**

- Compostie Primary Key, Foreign Key (TRANSACTION_ID +, PRODUCT_ID + BATCH_ID), quantity_sold, selling_price.

**ALLERY**

- ALLERGY_ID (Primary Key), PATIENT_ID (FOREIGN KEY), allergy_name.

# ER DIAGRAM


USER
PK USER_ID
first_name
last_name
date_of_birth
role
phone
address
password
gender
country
reg_date
STOCK CONTROLLER
PK CONTROLLER_ID
FK USER_ID
PURCHASE ORDER
PK ORDER_ID
FK SUPPLIER_ID
order_date
delivery_date
status
STOCK BATCH
PK BATCH_ID
FK PRODUCT_ID
batch_number
manuf_date
expiry_date
storage_conditions
quantity
unit_cost
FK CONTROLLER_ID
recieved_date
PHARMACIST
PK PHARMACIST_ID
FK USER_ID
license number
status
PURCHASE ORDER LINE
PK,FK1 ORDER_ID
PK,FK2 PRODUCT_ID
quantity_ordered
unit_cost
PRODUCT
PK PRODUCT_ID
product_name
description
dosage
category
price
FK SUPPLIER_ID
reorder_qty
storage_req
PATIENT
PK PATIENT_ID
FK USER_ID
passport
SUPPLIER
PK SUPPLIER_ID
supplier_name
email
address
status
payment_terms
phone
registration_number
PRESCRIPTION LINE
PK,FK PRESCRIPTION_ID
PK,FK PRODUCT_ID
quantity
dosage_description
PRESCRIPTION
PK PRESCRIPTION_ID
FK PATIENT_ID
FK PHARMACIST_ID
doctor_name
clinic/hospital_name
date_issued
ALLERGY
PK ALLERGY_ID
FK PATIENT_ID
allergy_name
SALES LINE
PK,FK TRANACTION_ID
PK,FK PRODUCT_ID
PK,FK BATCH_ID
quantity_sold
selling_price
CONTROLLED_LOG
PK LOG_ID
FK PRODUCT_ID
FK PHARMACIST_ID
FK BATCH_ID
FK PATIENT_ID
log_date
status
action_type
quantity
TRANSACTION
PK TRANSACTION_ID
transation_date
FK PharmacistID
FK PATIENT_ID
trans_type

# RELATIONSHIPS

1. **SUPPLIER AND PURCHASE ORDER**: a supplier can receive many purchase orders, but each purchase order can be handled by one supplier.
   **Cardinality**: 1:M – ONE TO MANY
   **Participation**: optional for a supplier to have purchase orders but mandatory for purchase orders to be handled by one supplier.

2. **PURCHASE ORDER AND PRODUCT**: a purchase order can have many orders of different products in it and each product can receive many purchases orders
   **Cardinality**: M:N – MANY TO MANY
   **Participation**: optional for product to have purchase orders but mandatory for purchase order to have a product ordered.

3. **PRODUCT AND STOCK BATCH**: a product can have many stock batches, but stock batch can consist of one product
   **Cardinality**: 1:M – ONE TO MANY
   **Participation**: it's both mandatory for a product to be in stock batches and stock batch to have the product

4. **PATIENT AND PRESCRIPTION**: a patient can have multiple prescriptions but only one prescription can belong to one patient,
   **Cardinality**: 1:M – ONE TO MANY
   **Participation**: both mandatory for a patient to have prescription and for prescription to belong to one patient.

5. **PHARMACIST AND PRESCRIPTION**: A pharmacist can handle many prescriptions, and each prescription can be handled by one pharmacist**.**
   **Cardinality**: 1:M – ONE TO MANY
   **Participation**: optional for a pharmacist to handle many prescriptions and it is mandatory for prescription to be handled by one pharmacist.

6. **PRESCRIPTION AND PRODUCT**: a prescription can have many products and products can have many prescriptions
   **Cardinality**: M:N – MANY TO MANY
   **Participation**: mandatory for a prescription to have products in it but optional for product to have a prescription.

7. **TRANSATION AND PRODUCT**: A product can have multiple transactions, and a transaction can have multiple products.
   **Cardinality**: M:N – MANY TO MANY
   **Participation:** It is mandatory for a transaction to include multiple products, but optional for a product to have a transaction.

8. **SUPPLIER AND PRODUCT:** A supplier can supply many products, but a product can only be supplied by one supplier.
   **Cardinality:** 1:M – ONE TO MANY
   **Participation:** It is optional for a supplier to supply many products, but it is mandatory for a product to have exactly one supplier.

9. **STOCK BATCH AND STOCK CONTROLLER:** A stock controller can control many stock batches, but stock batches can only be controlled by one stock controller.
   **Cardinality:** 1:M – ONE TO MANY
   **Participation**: It is mandatory for stock batches to be controlled by one stock controller, but optional for a stock controller to control many stock batches.

10. **PHARMACIST AND CONTROLLER LOG:** A pharmacist can generate many logs, and each log must belong to one pharmacist.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is mandatory for a generated log to belong to one pharmacist, but optional for a pharmacist to generate any log.

11. **TRANSACTION AND PATIENT:** A patient can have many transactions, but a transaction can only belong to one patient.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is optional for a transaction to belong to any patient, but mandatory for a patient to have a transaction.

12. **PHARMACIST AND TRANSACTION:** A pharmacist can process many transactions, but a transaction can only be processed by a pharmacist.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is optional for a transaction to be processed by a pharmacist, but mandatory for a pharmacist to process a transaction.

13. **PHARMACIST AND PATIENT:** A pharmacist can have multiple patients, and a patient can have multiple pharmacists.

**Cardinality:** M:N – MANY TO MANY
**Participation:** It is optional for a pharmacist to have a patient, but mandatory for a patient to have a pharmacist.

14. **PRODUCT AND CONTROLLED LOG**: A log can have multiple products, but a product can have only one log.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is optional for a product to have a controlled log, but mandatory for a control log to be linked to only one product.

15. **PRESCRIPTION AND PRESCRIPTION LINE**: A prescription can have many prescription lines, but a prescription line can exist in one prescription.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is mandatory for a prescription to have a prescription line and for a prescription line to belong to one prescription.

16. **PRODUCT AND PRESCRIPTION LINE:** A product can have multiple prescription lines, but a prescription line can only have one product.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation:** It is optional for a product to have prescription lines, but mandatory for a prescription line to be linked to one product.

17. **TRANSACTION AND SALES LINE:** Transactions can have multiple sales line, but sales line can only belong to one transaction.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation**: It is mandatory for a transaction to have at least one sales line, and a sales line must belong to exactly one transaction.

18. **PRODUCT AND SALES LINE:** A product can have multiple sales line, but a sales line can have one product.
    **Cardinality:** 1:M – ONE TO MANY
    **Participation**: It is optional for a product to have sales lines, but mandatory for a sales line to have one product.

19. **USER AND PHARMACIST:** A user can be a pharmacist
    **Cardinality:** 1:1 – ONE TO ONE
    **Participation:** It is optional for a user to be a pharmacist, but mandatory for a pharmacist to be a user.

20. **USER AND STOCK CONTROLLER**: A user can be a stock controller
    **Cardinality:** 1:1 – ONE TO ONE
    **Participation:** It is optional for a user to be a stock controller, but mandatory for a stock controller to be a user.

21. **USER AND PAT IENT:** A user can be a patient.
    **Cardinality:** 1:1 – ONE TO ONE
    **Participation:** It is optional for a user to be a patient, but mandatory for a patient to be a user.

# DIAGRAM NORMALIZATION

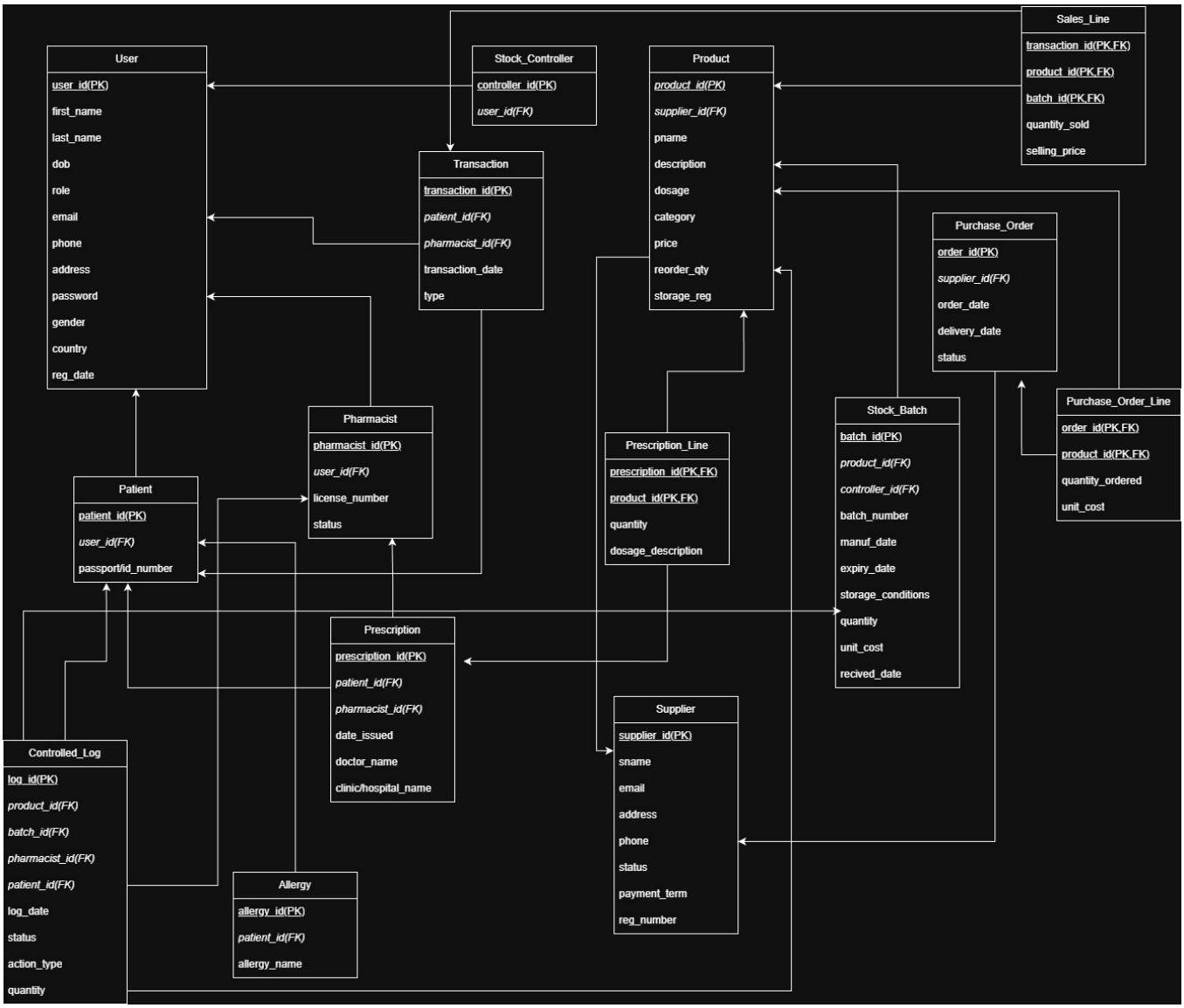
**NORMALAZATION CHECK**

| Normal Form | Requirement | Status | Explanation |
|---|---|---|---|
| 1NF | No repeating groups; atomic values | PASS | No multivalued attributes stored as lists |
| 2NF | No partial dependencies (for composite keys) | PASS | PRESCRIPTION_LINE, PURCHASE ORDER LIINE and SALES_LINE use composite keys correctly |
| 3NF | No transitive dependencies | PASS | No non-key attribute depends on another non-key attribute |

# FEATURES

| Feature | Location in Diagram | Explanation |
|---|---|---|
| Optional/Mandatory Relationships | Relationships 1, 5, 8, 9, 10 and 11 | Clearly indicated in relationship descriptions - "optional; supplier to purchase; mandatory; purchase order to supplier" |
| Weak Entities | PRESCRIPTION_LINE, SALES_LINE, PURCHASE ORDER LINE | cannot exist without parent PRESCRIPTION or TRANSACTION |
| Composite/Bridge Entities | PRESCRIPTION_LINE, SALES_LINE, PURCHASE ORDER LINE | Resolve M:N relationships between PRESCRIPTION-PRODUCT and TRANSACTION-PRODUCT and PRODUCT-PURCHASE ORDER |
| Composite Keys | PRESCRIPTION_LINE (PRESCRIPTION_ID + PRODUCT_ID), SALES_LINE (TRANSACTION_ID + PRODUCT_ID + BATCH_ID),PURCHASE ORDER LINE(ORDER_ID + PRODUCT_ID) | Primary keys made of multiple attributes |
| Composite Attributes | USER (first_name + last_name) | Two attributes that together form a full name |
| Multivalued Attributes | PATIENT (allergies) | A patient can have no, one or multiple allergies |
| Derived Attributes | USER (age) | Age is calculated from date of birth (dob) – not stored, only derived when needed. |

# LOGICAL DATA MODEL




**USER** (**<u>USER_ID</u>**, FIRST_NAME, LAST_NAME, DATE_OF _BIRTH, ROLE)

PRIMARY KEY: <u>USER_ID</u>

**PATIENT** (**PATIENT_ID**, FIRST_NAME, LAST_NAME, DATE_OF_BIRTH, CONTACT_NUMBER, EMAIL)

PRIMARY KEY: PATIENT_ID

**PATIENT_ALLERGY** (**ALLERGY_DESCRIPTION, PATIENT_ID**)

PRIMARY KEYS: ALLERGY_DESCRIPTION
PATIENT_ID

FOREIGN KEY: *PATIENT_ID REFERENCES PATIENT*

**SUPPLIER** (**SUPPLIER_ID**, SUPPLIER_NAME, CONTACT_PERSON, PHONE_NUMBER)

PRIMARY KEY: SUPPLIER_ID

**PRODUCT** (**PRODUCT_ID**, PRODUCT_NAME, DESCRIPTION, CATEGORY, REORDER_LEVEL)

PRIMARY KEY: PRODUCT_ID

**BATCH** (**BATCH_ID**, PRODUCT_ID, EXPIRY_DATE, QUANTITY_ON_HAND, COST_PRICE)

PRIMARY KEY: BATCH_ID

FOREIGN KEY: *PRODUCT_ID REFERENCES PRODUCT*

**PRESCRIPTION** (**PRESCRIPTION_ID**, PATIENT_ID, DOCTOR_NAME, DATE_ISSUED, STATUS)

PRIMARY KEY: PRESCRIPTION_ID

FOREIGN KEY**:** *PATIENT_ID REFERENCES PATIENT*

**PRESCRIPTION_LINE** (**PRESCRIPTION_ID**, **PRODUCT_ID**, DOSAGE_INSTRUCTIONS, QUANTITY_PRESCRIBED)

PRIMARY KEYS: PRESCRIPTION_ID
PRODUCT_ID

FOREIGN KEYS: *PRESCRIPTION_ID REFERENCES PRESCRIPTION*

*PRODUCT_ID REFERENCES PRODUCT*

**TRANSACTION** (**TRANSACTION_ID**, PATIENT_ID, USER_ID, TRANSACTION_DATE, TOTAL_AMOUNT)

PRIMARY KEY: TRANSACTION_ID

FOREIGN KEYS: *PATIENT_ID REFERENCES PATIENT*

*USER_ID REFERENCES USER*

**SALES_LINE** (**TRANSACTION_ID**, **PRODUCT_ID**, **BATCH_ID**, QUANTITY_SOLD, SELLING_PRICE)

PRIMARY KEYS: TRANSACTION_ID
PRODUCT_ID
BATCH_ID

FOREIGN KEYS: *TRANSACTION_ID REFERENCES TRANSACTION*

*PRODUCT_ID REFERENCES PRODUCT*

*BATCH_ID REFERENCES BATCH*

**PURCHASE_ORDER** (**ORDER_ID**, SUPPLIER_ID, ORDER_DATE, STATUS)

PRIMARY KEY: ORDER_ID

FOREIGN KEY: *SUPPLIER_ID REFERENCES SUPPLIER*

**PURCHASE_ORDER_LINE** (**ORDER_ID**, **PRODUCT_ID**, QUANTITY_ORDERED, UNIT_COST)

PRIMARY KEYS: ORDER_ID
PRODUCT_ID

FOREIGN KEYS: *ORDER_ID REFERENCES PURCHASE_ORDER*

*PRODUCT_ID REFERENCES PRODUCT*